

# Design Thinking with Generative AI

Powerful ChatGPT Skills for UX

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## Abstract

Generative AI is transforming UX design, and this course expands your design toolkit to leverage ChatGPT and DALL-E to save time and elevate your end-to-end design process. Classic methods like personas gain power. New approaches like “design reasoning at scale” automate UX workflows and use data pipelines, while “design reasoning in depth” applies new narrative methods. Learn prompting shorthand and prompt patterns. Explore simulation and time-saving UX research tools, and discover new ideation methods. Filter, score, select, and refine ideas. Apply AI to interaction and visual design while understanding ChatGPT’s limitations and pushing its boundaries.

## CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; Human computer interaction (HCI); Interaction paradigms; Interaction design; Interaction design process and methods; User centered design.

## Keywords

ChatGPT, LLMs, prompt engineering, design thinking

### ACM Reference Format:

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## 1 Benefits

Attendees will become proficient in using ChatGPT to solve design problems, to save time, and to raise the quality of their designs, while making delightful discoveries with ChatGPT. Learning to use ChatGPT for design means learning to think and reason with ChatGPT as a companion tool to amplify your reasoning skills.

Gaining skill with AI tools is paramount, as a 2024 Upwork study found that “47% of employees using AI say they have no idea how to achieve the productivity gains their employers expect” [1]. This course will provide prompting strategies, common prompting patterns, and a collection of useful prompts specifically tuned for UX tasks in design, ideation, and user research. We’ll cover ChatGPT

shortcuts that move the conversation quickly to the needed results instead of through complicated prompt constructions.

Attendees will learn an end-to-end design approach that provides value in each stage of UX, and meta-prompts that organize across the UX workflow, such as the example in Figure 1, triggering multiple design steps in a single prompt. We’ll look at how to collect and manipulate the data and ideas you collect, and how to process them through a data pipeline toward final design decisions. You’ll learn interesting and useful prompts for personas, ideation, user research, requirements, ethical considerations, interaction, and visual design.

## 2 Intended Audience(s)

This course is for anyone looking to get more out of their work with ChatGPT in applying it to problems in HCI/UX. UX practitioners, students, and researchers will all benefit.

## 3 Prerequisites

A basic knowledge of ChatGPT is all that’s needed: a personal account and experience trying a few design prompts. We’ll take it from there. At the time of this writing, those wishing to generate more than a few images and access websites, which is recommended, will need the paid ChatGPT Plus account, which also produces somewhat better reasoning, and thus more satisfying results.

A basic knowledge of the UX design process is helpful, as we will dive into looking at ways to facilitate common steps of the process. Any HCI foundations course at the conference will provide what you need. ChatGPT can fill in gaps in your knowledge.

## 4 Content

The goal of this course is to make you better at UX and design in general by having you bring ChatGPT into the equation as a partner in design thinking. We’ll introduce prompting techniques (prompt engineering), focusing on efficient and effective prompts, and talk about how you reason with information in ChatGPT with two related techniques: workflows and data pipelines. Data pipelines transform data systematically into forms needed for each design stage. We’ll cover powerful things you can do with a table, like translate lists of pain points into features, and translate those into design suggestions. All of this acknowledges both the unique benefits of ChatGPT and the limitations, even problems, of working in a non-deterministic AI environment that can’t always provide predictable results. Then we’ll work through a wide range of UX methods, where we’ll understand each in terms of the pain points each addresses in the design process.

Topics to be covered are broken into these two 75-minute segments:

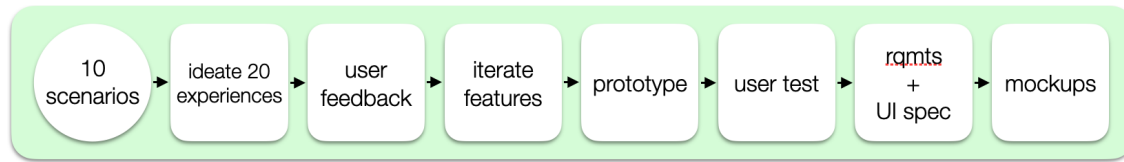
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**Re: Gamified language learning mobile app**

"telegraphic interim results lists, step by step, 1 per prompt, y/n to continue"

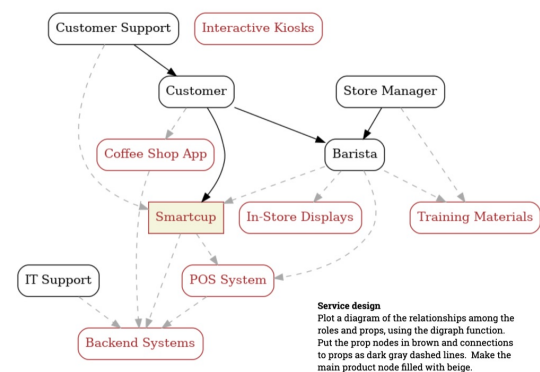
**Figure 1:** A workflow prompt includes a topic (a gamified language learning mobile app), the steps of the workflow, and a description of the style for conducting the workflow (writing style, type of detail, and process for stepping through the workflow).



**Figure 2:** Three persona images for a smart cup concept. If we recognize that lo-fidelity images serve a purpose in various aspects of our process, the imperfections of image generation become acceptable due to the strengths of the AI-generated approach. In the case of persona images, some of the strengths include: personas with unique actions, props, and environmental contexts, as well as flexibility in the style of representing them. (This image was generated with ChatGPT.)

#### •ChatGPT Techniques

- Quick intro to ChatGPT
- Non-determinism: ideas and reasoning, but not truth. Knowledge & sources/fact limits.
- Prompting skills (UX-motivated), shorthand, prompt patterns (design oriented), variables
- Workflows and problem solving
- The whole UX process and the AI/human handoff (who takes what role?)
- Non-deterministic data pipelines and table techniques
- Prompt chaining/refinement/fixing/clarifying/conversation, summarizing/clustering
- Ideation as a core skill of design thinking, with more ideas than ever



**Figure 3:** This diagram from ChatGPT maps out the relationships between roles and props in a service design scenario for a smartcup concept being sold through a coffee chain. ChatGPT writes and executes the Python code to generate a custom plot.

#### •Design & User Research

- Requirements and competitive analysis
- Audience and personas (Figure 2)
- Rating, scoring, traffic lights, and gold stars
- User research planning, support, and analysis
- Structured visuals with charts and HTML output (Figure 3)
- Information architecture and interaction design
- Visual design and visualization with Dall-E, including using imperfect visuals for low-fidelity mockups

This course draws from a university course on the same topic, leveraging ChatGPT to transform the way that design thinking and UX is practiced. The course slides will be made available on a course website, with a prompt reference guide for all prompts covered in the slides, as well as related prompts.

## 5 Practical work

We will devote 20-30 min of each of the sessions to prompt exercises in ChatGPT. If internet access is limited, I will have backup examples and additional Q&A time. Participants should bring laptops or may

also use the ChatGPT app on their phones or tablets. Pair work is encouraged to help with difficulties.

Students should choose a sample design concept they'd like to develop, and may bring ideas from home—it should not be anything confidential, as anything entered may be used for training of ChatGPT. As examples, common design concepts I work with are: a touchscreen smart cup, a hotel kiosk, and a gamified language learning app. In each exercise, you'll be given a few prompts to try with your concept, and then several optional prompts to choose your direction and play with the possibilities.

## 6 Instructor background

Tom Brinck is a UX designer and educator with over 35 years in the industry, including roles at Cisco, Samsung, and Amazon. He has led multidisciplinary teams in consumer electronics, telecommunications, and web design, with a focus on usability and innovation. He previously co-founded a UX and web design agency, where he worked with major clients across various industries, and taught graduate courses on UX & UI design at both the University of Michigan and Academy of Art University. Tom now focuses on generative AI's potential in UX. He is an adjunct instructor at the

University of Maryland, integrating ChatGPT methods into each HCI course, including a Fall 2024 course on "Generative AI in UX: Transforming UX Practice". Tom's publications include the book "Usability for the Web", which covers web design best practices. He is passionate about empowering designers to use new technologies effectively, balancing creativity, usability, and ethics.

## 7 Accessibility

All presentation slides will be designed for accessibility, and I will work to support attendees with any special needs.

## Acknowledgments

I've really appreciated the efforts and feedback of students at the University of Maryland who have tried out UX methods within ChatGPT in three courses in Inclusive Design, Advanced Usability Testing, and Generative AI in UX.

## References

- [1] Kelly Monahan and Gabby Burlacu. From Burnout to Balance: AI-Enhanced Work Models. Retrieved Oct 10, 2024 from <https://www.upwork.com/research/ai-enhanced-work-models>